

Identidades de sumatorias y logaritmos

$$\sum_{i=1}^n c = nc \quad (1.5.1)$$

$$\sum_{i=k}^n c = (n - k + 1)c \quad (1.5.2)$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \quad (1.5.3)$$

$$\sum_{i=0}^n i^2 = \frac{n(n+1)(2n+1)}{6} \quad (1.5.4)$$

$$\sum_{i=0}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2 \quad (1.5.5)$$

$$\sum_{i=0}^n i^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} \quad (1.5.6)$$

$$\sum_{i=k}^n f(i) = \sum_{i=1}^n f(i) - \sum_{i=1}^{k-1} f(i) \quad (1.5.7)$$

$$\sum_{i=0}^n 2^i = 2^{n+1} - 1 \quad (1.5.8)$$

$$\sum_{i=0}^n a^i = \frac{1 - a^{n+1}}{1 - a} \quad (1.5.9)$$

$$\log_b a = \frac{\log_c a}{\log_c b} \quad (1.5.10)$$

$$\log \frac{a}{b} = \log a - \log b \quad (1.5.11)$$

$$\log(a \times b) = \log a + \log b \quad (1.5.12)$$

$$\log_b(b^a) = b^{\log_b a} = a \quad (1.5.13)$$

$$(a^b)^c = a^{bc} = (a^c)^b \quad (1.5.14)$$

$$\sum_{i=m}^n i = \frac{n(n+1)}{2} - \frac{m(m-1)}{2}$$